

ACEest Fitness & Gym DevOps Implementation Report

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1. PROJECT OVERVIEW

This project implements a complete DevOps workflow for the ACEest Fitness & Gym Flask application. The workflow integrates GitHub Flow, pull request validation, GitHub Actions, Jenkins CI/CD, Sonar CLI static analysis, Pytest/Selenium validation, Docker image creation, Dockerhub container Registry publishing, and Kubernetes progressive deployment strategies for Assignment 2.

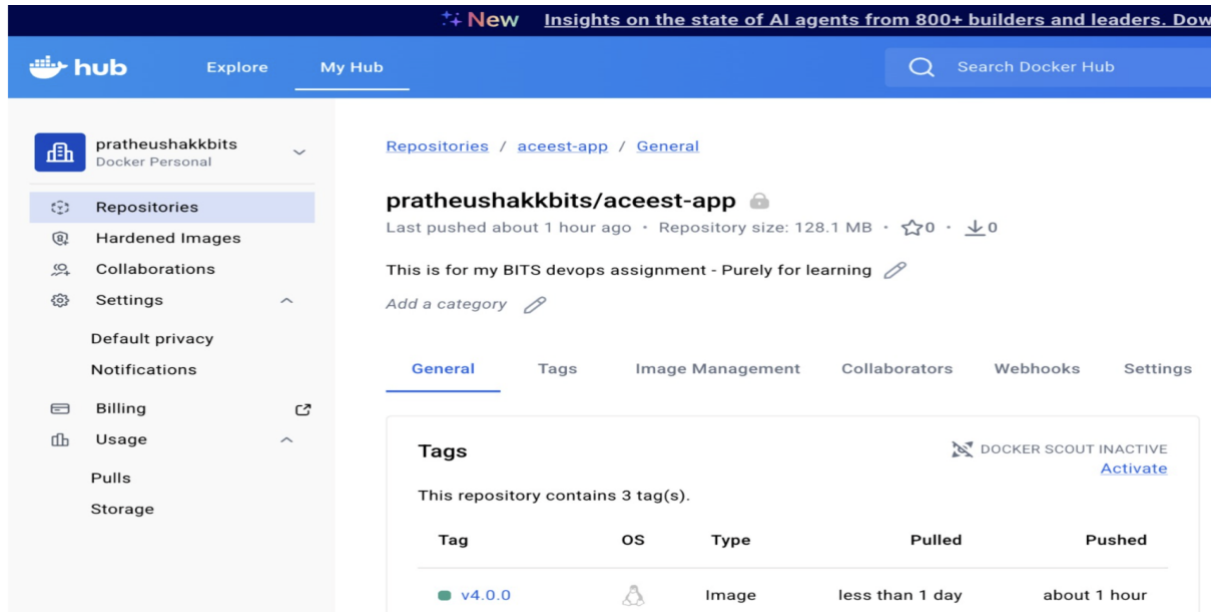
2. APPLICATION SUMMARY

- Technology stack: Flask, Python, Pytest, Selenium/Behave, GitHub Actions, Jenkins, Sonar CLI, Docker, Docker Hub container Registry, Kubernetes, and Minikube.
- Main features: REST APIs and GUI pages for the health endpoint, program listing, program details, and calorie estimation.
- The application codebase and CI/CD practices were reused from Assignment 1 and extended with Kubernetes manifests and rollout strategy evidence for Assignment 2.

3. CI/CD ARCHITECTURE

- Developers use GitHub Flow with short-lived feature branches and pull requests into the production-ready main branch.
- GitHub Actions runs repository-native CI on push and pull_request events, while Jenkins uses Poll SCM because webhooks were not reliable in the Azure virtual lab network.
- Pytest validates Flask endpoints such as /, /health, /programs,/programs/<program_name>, and /estimate-calories before packaging.
- Sonar CLI performs static analysis from Jenkins using sonar-project.properties and archives the generated report.
- Docker builds a secure multi-stage image and Jenkins pushes aceest-app images to Docker Hub container Registry

```
07:12:48 Successfully built 97edd566dde9
07:12:48 Successfully tagged aceest-app:v4.0.0
07:12:48 + docker tag aceest-app:v4.0.0 pratheushakkbites/aceest-app:v4.0.0
07:12:48 + docker image ls aceest-app:v4.0.0 pratheushakkbites/aceest-app:v4.0.0
```

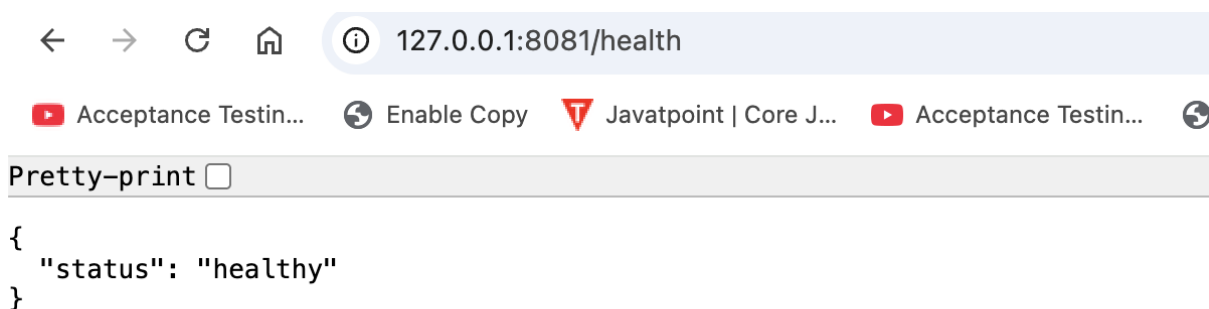
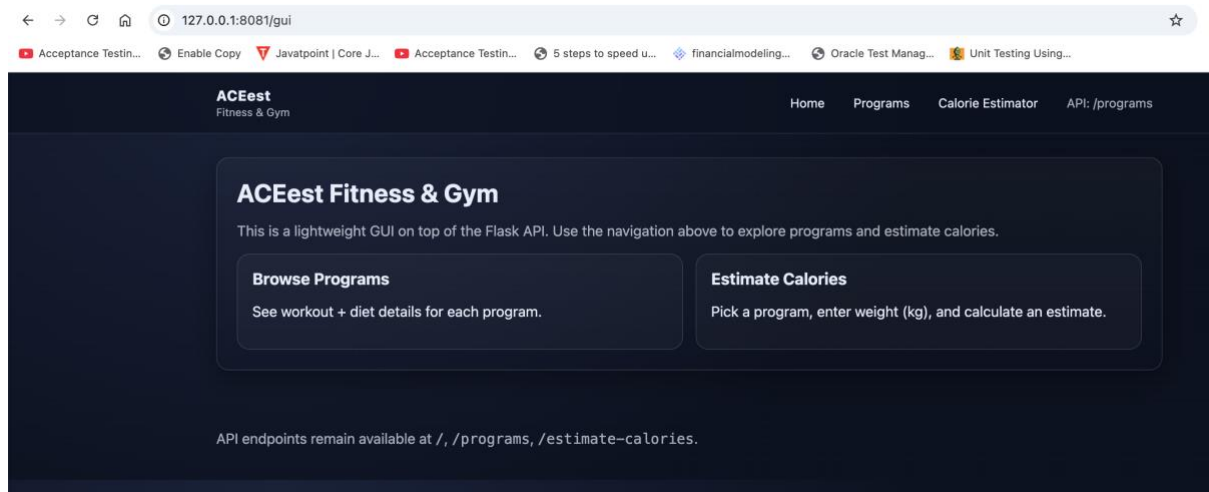


- Kubernetes/Minikube deploys the image and validates rolling update, blue-green, canary, shadow, and A/B testing strategies.
- Version Control and Branching: The Assignment 1 implementation was maintained in the main branch, while the Assignment 2 changes, Kubernetes assets, Jenkins local setup, deployment evidence, and report-related assets were developed under the [assignment-assets](#) branch. This separation helped preserve the original Assignment 1 baseline while allowing Assignment 2 DevOps enhancements to be tracked independently.

4. PIPELINE STAGES IMPLEMENTED

- Checkout source code from the GitHub repository: <https://github.com/Pratheusha-kk/aceest-app/tree/main>.
- Run Sonar CLI static analysis in Jenkins and archive the report output.
- Execute Pytest unit/integration tests and optional Selenium/Behave UI tests.
- Build the multi-stage Docker image and smoke-test /, /health, /programs, and /gui.
- Push versioned aceest-app images to Docker Hub container Registry - Docker Hub Repository: pratheushakkbits/aceest-app.
- Deploy the image to Kubernetes/Minikube and validate the progressive rollout strategy evidence.

```
Last login: Thu Apr 30 10:19:50 on ttys001
pratheushakk@Pratheushas-MacBook-Pro ~ % k get nodes
NAME      STATUS   ROLES    AGE   VERSION
minikube  Ready    control-plane  29h   v1.35.1
pratheushakk@Pratheushas-MacBook-Pro ~ % k get pods -n aceest
NAME                                READY   STATUS    RESTARTS   AGE
aceest-app-f79c4957b-5qbst           1/1     Running   0           56m
aceest-app-f79c4957b-l7gxr           1/1     Running   0           56m
aceest-app-f79c4957b-zvbgd           1/1     Running   0           56m
pratheushakk@Pratheushas-MacBook-Pro ~ % k port-forward -n aceest svc/aceest-app 8081:80
Forwarding from 127.0.0.1:8081 -> 5000
Forwarding from [::1]:8081 -> 5000
```



5. TESTING AND QUALITY VALIDATION

- Pytest result: Unit and integration tests passed for API health, program catalog, program details, calorie estimation, and invalid input handling.

pytest-report.html

Report generated on 30-Apr-2026 at 02:26:20 by [pytest-html](#) v4.2.0

Environment

Summary

8 tests took 00:00:04.

(Un)check the boxes to filter the results.

There are still tests running.

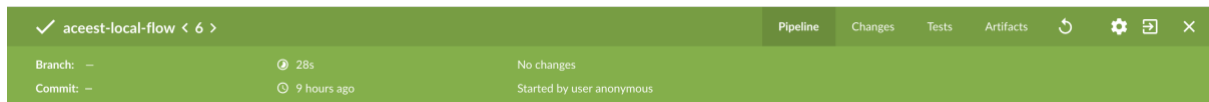
Reload this page to get the latest results!

☐ 0 Failed, ☒ 8 Passed, ☐ 0 Skipped, ☐ 0 Expected failures, ☐ 0 Unexpected passes, ☐ 0 Errors, ☐ 0 Reruns ☐ 0 Retried,

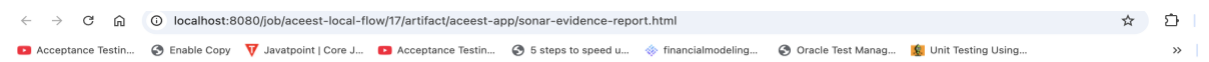
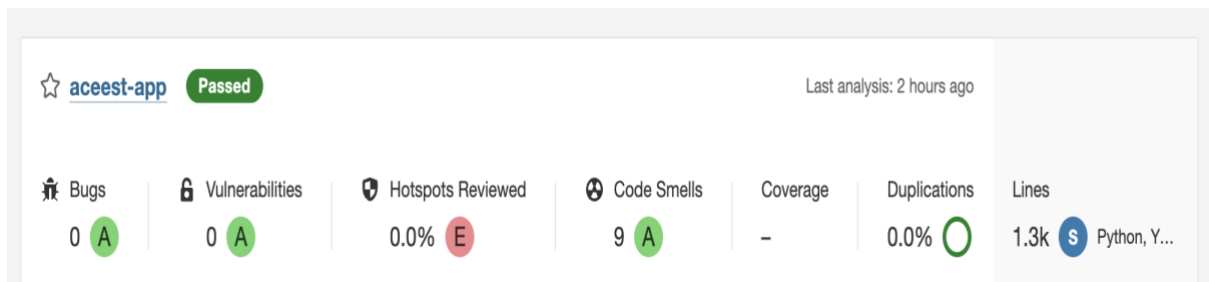
[Show all details](#) / [Hide all details](#)

Result Test Duration Links

- Jenkins result: Pipeline executes checkout, SonarQube – Static Analysis , Pytest , UI tests, Docker build, Docker Hub image push , minikube deployment stages.



- Sonar result: The SonarQube limitation from the earlier setup was rectified by hosting SonarQube locally at <http://localhost:9000>. Jenkins was configured to execute SonarScanner as part of the CI/CD pipeline, using the project's `sonar-project.properties` file for static code analysis configuration. After the scan completed, the SonarQube reports and related artifacts were archived in Jenkins, providing evidence of code quality validation as part of the automated pipeline.



ACEest Local SonarScanner Evidence Report

Generated at 2026-04-30T01:16:19.480918+00:00.

This report accompanies `sonar-report.txt`. It confirms the real SonarScanner CLI ran locally and summarizes the code scanned. A full SonarQube dashboard and quality gate require `SONAR_HOST_URL` and `SONAR_TO`

Summary

7Python files
964Total lines
781Code lines
0Syntax failures

Sonar Project Properties

- `sonar.exclusions`: generated-docs/**.docs/**.docx
- `sonar.host.url`: <http://sonarqube:9000>
- `sonar.login`: `squ_60630e24e2c56c6e8523563ec7279fb8a155bd77`
- `sonar.projectKey`: `aceest-app`
- `sonar.projectName`: `aceest-app`
- `sonar.projectVersion`: `1.0`
- `sonar.python.version`: `3.11`
- `sonar.sources`: `app.py,scripts,tests,ui-tests.k8s`

Python Static Check Inventory

File	Lines	Code	Comments	Functions	Classes	Syntax
app.py	169	138	0	10	0	PASS
scripts/create_assignment2_template.py	279	238	0	8	0	PASS
scripts/generate_sonar_evidence_report.py	230	195	1	7	1	PASS
tests/test_app.py	58	44	0	6	1	PASS
tests/test_integration.py	126	101	4	6	1	PASS
ui-tests/features/environment.py	36	25	2	2	0	PASS
ui-tests/features/steps/tabs_steps.py	66	40	8	8	0	PASS

- Docker image tags created: `aceest-app` images published to Docker Hub container Registry using `branch/build` tags; Kubernetes strategy validation uses `stable v4.0.0` and `candidate v4.0.1` tags.

6. MINIKUBE SETTINGS IN LOCAL

```
pratheushakk@Pratheushas-MacBook-Pro aceest-app % kubectl config use-context minikube
kubectl get nodes
minikube status

Switched to context "minikube".
NAME      STATUS   ROLES    AGE   VERSION
minikube   Ready    control-plane  3m50s  v1.35.1
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured

pratheushakk@Pratheushas-MacBook-Pro aceest-app % kubectl config use-context minikube
Switched to context "minikube".
pratheushakk@Pratheushas-MacBook-Pro aceest-app % kubectl get nodes
NAME      STATUS   ROLES    AGE   VERSION
minikube   Ready    control-plane  4m22s  v1.35.1
pratheushakk@Pratheushas-MacBook-Pro aceest-app % minikube status
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured
```

```
pratheushakk@Pratheushas-MacBook-Pro aceest-app % docker images | grep aceest-app

WARNING: This output is designed for human readability. For machine-readable output, please use --format.
aceest-app:v3.1.2          5517fb8b0290      225MB      48.4MB
aceest-app:v3.2.4          eb4b1b5e5778      225MB      48.4MB
```

7. DEPLOYMENT AND ROLLBACK STRATEGY EVIDENCE

The following evidence blocks summarize the rollout strategies validated with Kubernetes/Minikube commands and terminal outputs.

7.1 ROLLING UPDATE

```
pratheushakk@Pratheushas-MacBook-Pro aceest-app % kubectl annotate deployment/aceest-app -n aceest \
  kubernetes.io/change-cause="Initial deploy with stable image aceest-app:v4.0.0" --overwrite

deployment.apps/aceest-app annotated
pratheushakk@Pratheushas-MacBook-Pro aceest-app % kubectl set image deployment/aceest-app aceest-app=aceest-app:v4.0.0 -n aceest
kubectl rollout status deployment/aceest-app -n aceest

deployment "aceest-app" successfully rolled out
pratheushakk@Pratheushas-MacBook-Pro aceest-app % kubectl get deploy,rs,pods,svc -n aceest -o wide
```

NAME	READY	UP-TO-DATE	AVAILABLE	AGE	CONTAINERS	IMAGES	SELECTOR
deployment.apps/aceest-app	3/3	3	3	61s	aceest-app	aceest-app:v4.0.0	app=aceest-app,strategy=rolling-update

NAME	DESIRED	CURRENT	READY	AGE	CONTAINERS	IMAGES	SELECTOR
replicaset.apps/aceest-app-66889d8d58	0	0	0	61s	aceest-app	aceest-app:latest	app=aceest-app,pod-template-hash=66889d8d58,strategy=rolling-update
replicaset.apps/aceest-app-7f779d54f9	3	3	3	61s	aceest-app	aceest-app:v4.0.0	app=aceest-app,pod-template-hash=7f779d54f9,strategy=rolling-update

NAME	READY	STATUS	RESTARTS	AGE	IP	NODE	NOMINATED NODE	READINESS GATES
pod/aceest-app-7f779d54f9-bpc4f	1/1	Running	0	54s	10.244.0.46	minikube	<none>	<none>
pod/aceest-app-7f779d54f9-jrncw	1/1	Running	0	46s	10.244.0.47	minikube	<none>	<none>
pod/aceest-app-7f779d54f9-k7nf8	1/1	Running	0	61s	10.244.0.45	minikube	<none>	<none>

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)	AGE	SELECTOR
service/aceest-app	ClusterIP	10.110.111.212	<none>	80/TCP	61s	app=aceest-app

```
pratheushakk@Pratheushas-MacBook-Pro aceest-app % kubectl annotate deployment/aceest-app -n aceest \
  kubernetes.io/change-cause="Rolling update to candidate image aceest-app:v4.0.1" --overwrite

deployment.apps/aceest-app annotated
pratheushakk@Pratheushas-MacBook-Pro aceest-app % kubectl set image deployment/aceest-app aceest-app=aceest-app:v4.0.1 -n aceest
kubectl rollout status deployment/aceest-app -n aceest

deployment.apps/aceest-app image updated
Waiting for deployment "aceest-app" rollout to finish: 1 out of 3 new replicas have been updated...
Waiting for deployment "aceest-app" rollout to finish: 1 out of 3 new replicas have been updated...
Waiting for deployment "aceest-app" rollout to finish: 1 out of 3 new replicas have been updated...
Waiting for deployment "aceest-app" rollout to finish: 2 out of 3 new replicas have been updated...
Waiting for deployment "aceest-app" rollout to finish: 2 out of 3 new replicas have been updated...
Waiting for deployment "aceest-app" rollout to finish: 2 out of 3 new replicas have been updated...
Waiting for deployment "aceest-app" rollout to finish: 2 out of 3 new replicas have been updated...
Waiting for deployment "aceest-app" rollout to finish: 2 out of 3 new replicas have been updated...
Waiting for deployment "aceest-app" rollout to finish: 1 old replicas are pending termination...
Waiting for deployment "aceest-app" rollout to finish: 1 old replicas are pending termination...
Waiting for deployment "aceest-app" rollout to finish: 1 old replicas are pending termination...
deployment "aceest-app" successfully rolled out
pratheushakk@Pratheushas-MacBook-Pro aceest-app % kubectl get deploy,rs,pods,svc -n aceest -o wide
```

NAME	READY	UP-TO-DATE	AVAILABLE	AGE	CONTAINERS	IMAGES	SELECTOR
deployment.apps/aceest-app	3/3	3	3	115s	aceest-app	aceest-app:v4.0.1	app=aceest-app,strategy=rolling-update

NAME	DESIRED	CURRENT	READY	AGE	CONTAINERS	IMAGES	SELECTOR
replicaset.apps/aceest-app-65d5f69cb4	3	3	3	30s	aceest-app	aceest-app:v4.0.1	app=aceest-app,pod-template-hash=65d5f69cb4,strategy=rolling-update
replicaset.apps/aceest-app-66889d8d58	0	0	0	115s	aceest-app	aceest-app:latest	app=aceest-app,pod-template-hash=66889d8d58,strategy=rolling-update
replicaset.apps/aceest-app-7f779d54f9	0	0	0	115s	aceest-app	aceest-app:v4.0.0	app=aceest-app,pod-template-hash=7f779d54f9,strategy=rolling-update

NAME	READY	STATUS	RESTARTS	AGE	IP	NODE	NOMINATED NODE	READINESS GATES
pod/aceest-app-65d5f69cb4-jffsh	1/1	Running	0	30s	10.244.0.48	minikube	<none>	<none>
pod/aceest-app-65d5f69cb4-k8qng	1/1	Running	0	22s	10.244.0.49	minikube	<none>	<none>
pod/aceest-app-65d5f69cb4-rsjhw	1/1	Running	0	15s	10.244.0.50	minikube	<none>	<none>

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)	AGE	SELECTOR
service/aceest-app	ClusterIP	10.110.111.212	<none>	80/TCP	115s	app=aceest-app

```
pratheushakk@Pratheushas-MacBook-Pro aceest-app % kubectl rollout history deployment/aceest-app -n aceest

deployment.apps/aceest-app
REVISION  CHANGE-CAUSE
1          <none>
2          Rolling update to candidate image aceest-app:v4.0.1
3          Rolling update to candidate image aceest-app:v4.0.1
```

7.2 BLUE-GREEN DEPLOYMENT

```
pratheushakk@Pratheushas-MacBook-Pro aceest-app % kubectl apply -n aceest -f k8s/blue-green/service-active-green.yaml
kubectl describe svc aceest-app-active -n aceest

service/aceest-app-active configured
Name:                aceest-app-active
Namespace:           aceest
Labels:              <none>
Annotations:         <none>
Selector:            app=aceest-app,track=green
Type:                ClusterIP
IP Family Policy:    SingleStack
IP Families:         IPv4
IP:                  10.107.131.176
IPs:                 10.107.131.176
Port:                http 80/TCP
TargetPort:          5000/TCP
Endpoints:           10.244.0.58:5000,10.244.0.56:5000
Session Affinity:    None
Internal Traffic Policy: Cluster
Events:              <none>
pratheushakk@Pratheushas-MacBook-Pro aceest-app % kubectl rollout history deployment/aceest-app-blue -n aceest
kubectl rollout history deployment/aceest-app-green -n aceest

deployment.apps/aceest-app-blue
REVISION  CHANGE-CAUSE
1         <none>
2         Blue environment deployed with stable image aceest-app:v4.0.0

deployment.apps/aceest-app-green
REVISION  CHANGE-CAUSE
1         <none>
2         Green environment deployed with candidate image aceest-app:v4.0.1

pratheushakk@Pratheushas-MacBook-Pro aceest-app % kubectl annotate deployment/aceest-app-blue -n aceest \
  kubernetes.io/change-cause="Blue environment deployed with stable image aceest-app:v4.0.0" --overwrite
kubectl annotate deployment/aceest-app-green -n aceest \
  kubernetes.io/change-cause="Green environment deployed with candidate image aceest-app:v4.0.1" --overwrite

deployment.apps/aceest-app-blue annotated
deployment.apps/aceest-app-green annotated
pratheushakk@Pratheushas-MacBook-Pro aceest-app % kubectl set image deployment/aceest-app-blue aceest-app=aceest-app:v4.0.0 -n aceest
kubectl set image deployment/aceest-app-green aceest-app=aceest-app:v4.0.1 -n aceest
kubectl rollout status deployment/aceest-app-blue -n aceest
kubectl rollout status deployment/aceest-app-green -n aceest
kubectl get deploy,pods,svc -n aceest -o wide

deployment "aceest-app-blue" successfully rolled out
deployment "aceest-app-green" successfully rolled out

```

NAME	READY	UP-TO-DATE	AVAILABLE	AGE	CONTAINERS	IMAGES	SELECTOR
deployment.apps/aceest-app-blue	2/2	2	2	56s	aceest-app	aceest-app:v4.0.0	app=aceest-app,strategy=blue-green,track=blue
deployment.apps/aceest-app-green	2/2	2	2	56s	aceest-app	aceest-app:v4.0.1	app=aceest-app,strategy=blue-green,track=green

NAME	READY	STATUS	RESTARTS	AGE	IP	NODE	NOMINATED NODE	READINESS GATES
pod/aceest-app-blue-5cf84ccd7-9pgs5	1/1	Running	0	56s	10.244.0.55	minikube	<none>	<none>
pod/aceest-app-blue-5cf84ccd7-cn9k2	1/1	Running	0	55s	10.244.0.57	minikube	<none>	<none>
pod/aceest-app-green-67cb4d98ff-m0tsv	1/1	Running	0	56s	10.244.0.56	minikube	<none>	<none>
pod/aceest-app-green-67cb4d98ff-ngbwq	1/1	Running	0	55s	10.244.0.58	minikube	<none>	<none>

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)	AGE	SELECTOR
service/aceest-app-active	ClusterIP	10.107.131.176	<none>	80/TCP	56s	app=aceest-app,track=blue
service/aceest-app-preview	ClusterIP	10.100.142.92	<none>	80/TCP	56s	app=aceest-app,track=green

7.3 CANARY RELEASE

```
pratheushakk@Pratheushas-MacBook-Pro aceest-app % kubectl annotate deployment/aceest-app-stable -n aceest \
  kubernetes.io/change-cause="Stable canary baseline deployed with image aceest-app:v4.0.0" --overwrite
kubectl set image deployment/aceest-app-stable aceest-app=aceest-app:v4.0.0 -n aceest
kubectl rollout status deployment/aceest-app-stable -n aceest

deployment.apps/aceest-app-stable annotated
deployment "aceest-app-stable" successfully rolled out
```

```
pratheushakk@Pratheushas-MacBook-Pro aceest-app % kubectl annotate deployment/aceest-app-canary -n aceest \
  kubernetes.io/change-cause="Canary candidate deployed with image aceest-app:v4.0.1" --overwrite
kubectl set image deployment/aceest-app-canary aceest-app=aceest-app:v4.0.1 -n aceest
kubectl rollout status deployment/aceest-app-canary -n aceest
```

```
deployment.apps/aceest-app-canary annotated
deployment "aceest-app-canary" successfully rolled out
```

```
pratheushakk@Pratheushas-MacBook-Pro aceest-app % kubectl get deploy,pods,svc -n aceest -o wide
kubectl get pods -n aceest -L track,strategy
kubectl rollout history deployment/aceest-app-stable -n aceest
kubectl rollout history deployment/aceest-app-canary -n aceest
```

NAME	READY	UP-TO-DATE	AVAILABLE	AGE	CONTAINERS	IMAGES	SELECTOR
deployment.apps/aceest-app-canary	1/1	1	1	91s	aceest-app	aceest-app:v4.0.1	app=aceest-app,strategy=canary,track=canary
deployment.apps/aceest-app-stable	4/4	4	4	91s	aceest-app	aceest-app:v4.0.0	app=aceest-app,strategy=canary,track=stable

NAME	READY	STATUS	RESTARTS	AGE	IP	NODE	NOMINATED NODE	READINESS GATES
pod/aceest-app-canary-584d9fdbdc-w49tm	1/1	Running	0	91s	10.244.0.74	minikube	<none>	<none>
pod/aceest-app-stable-67766d84f8-7fg55	1/1	Running	0	89s	10.244.0.76	minikube	<none>	<none>
pod/aceest-app-stable-67766d84f8-rrl2d	1/1	Running	0	91s	10.244.0.73	minikube	<none>	<none>
pod/aceest-app-stable-67766d84f8-vxx5t	1/1	Running	0	91s	10.244.0.72	minikube	<none>	<none>
pod/aceest-app-stable-67766d84f8-zw9pr	1/1	Running	0	90s	10.244.0.75	minikube	<none>	<none>

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)	AGE	SELECTOR
service/aceest-app	ClusterIP	10.106.67.199	<none>	80/TCP	91s	app=aceest-app

NAME	READY	STATUS	RESTARTS	AGE	TRACK	STRATEGY
aceest-app-canary-584d9fdbdc-w49tm	1/1	Running	0	92s	canary	canary
aceest-app-stable-67766d84f8-7fg55	1/1	Running	0	90s	stable	canary
aceest-app-stable-67766d84f8-rrl2d	1/1	Running	0	92s	stable	canary
aceest-app-stable-67766d84f8-vxx5t	1/1	Running	0	92s	stable	canary
aceest-app-stable-67766d84f8-zw9pr	1/1	Running	0	91s	stable	canary

```
deployment.apps/aceest-app-stable
REVISION CHANGE-CAUSE
1 <none>
2 Stable canary baseline deployed with image aceest-app:v4.0.0
```

```
deployment.apps/aceest-app-canary
REVISION CHANGE-CAUSE
1 <none>
2 Canary candidate deployed with image aceest-app:v4.0.1
```

```
pratheushakk@Pratheushas-MacBook-Pro aceest-app % kubectl annotate deployment/aceest-app-canary -n aceest \
  kubernetes.io/change-cause="Canary rolled back by scaling candidate to zero" --overwrite
deployment.apps/aceest-app-canary annotated
pratheushakk@Pratheushas-MacBook-Pro aceest-app % kubectl scale deployment/aceest-app-canary --replicas=0 -n aceest
deployment.apps/aceest-app-canary scaled
```

NAME	READY	UP-TO-DATE	AVAILABLE	AGE	CONTAINERS	IMAGES	SELECTOR
deployment.apps/aceest-app-canary	0/0	0	0	7m	aceest-app	aceest-app:v4.0.1	app=aceest-app,strategy=canary,track=canary
deployment.apps/aceest-app-stable	4/4	4	4	7m	aceest-app	aceest-app:v4.0.0	app=aceest-app,strategy=canary,track=stable

NAME	READY	STATUS	RESTARTS	AGE	IP	NODE	NOMINATED NODE	READINESS GATES
pod/aceest-app-stable-67766d84f8-7fg55	1/1	Running	0	6m58s	10.244.0.76	minikube	<none>	<none>
pod/aceest-app-stable-67766d84f8-rrl2d	1/1	Running	0	7m	10.244.0.73	minikube	<none>	<none>
pod/aceest-app-stable-67766d84f8-vxx5t	1/1	Running	0	7m	10.244.0.72	minikube	<none>	<none>
pod/aceest-app-stable-67766d84f8-zw9pr	1/1	Running	0	6m59s	10.244.0.75	minikube	<none>	<none>

```
pratheushakk@Pratheushas-MacBook-Pro aceest-app % kubectl rollout history deployment/aceest-app-canary -n aceest
deployment.apps/aceest-app-canary
REVISION CHANGE-CAUSE
1 <none>
2 Canary rolled back by scaling candidate to zero
```

7.4 SHADOW DEPLOYMENT

```
pratheushakk@Pratheushas-MacBook-Pro aceest-app % kubectl annotate deployment/aceest-app-primary -n aceest \
  kubernetes.io/change-cause="Primary shadow environment deployed with stable image aceest-app:v4.0.0" --overwrite
```

```
kubectl annotate deployment/aceest-app-shadow -n aceest \
  kubernetes.io/change-cause="Shadow environment deployed with candidate image aceest-app:v4.0.1" --overwrite
```

```
deployment.apps/aceest-app-primary annotated
deployment.apps/aceest-app-shadow annotated
```

```

pratheushakk@Pratheushas-MacBook-Pro aceest-app % kubectl set image deployment/aceest-app-primary aceest-app=aceest-app:v4.0.0 -n aceest
kubectl set image deployment/aceest-app-shadow aceest-app=aceest-app:v4.0.1 -n aceest
kubectl rollout status deployment/aceest-app-primary -n aceest
kubectl rollout status deployment/aceest-app-shadow -n aceest
kubectl get deploy,pods,svc,ingress -n aceest -o wide
kubectl describe ingress aceest-app-shadow -n aceest

deployment "aceest-app-primary" successfully rolled out
deployment "aceest-app-shadow" successfully rolled out
NAME                                READY    UP-TO-DATE    AVAILABLE    AGE    CONTAINERS    IMAGES                                SELECTOR
deployment.apps/aceest-app-primary    2/2      2              2            103s   aceest-app    aceest-app:v4.0.0                    app=aceest-app-shadow,strategy=shadow,track=primary
deployment.apps/aceest-app-shadow     1/1      1              1            103s   aceest-app    aceest-app:v4.0.1                    app=aceest-app-shadow,strategy=shadow,track=shadow

NAME                                READY    STATUS    RESTARTS    AGE    IP            NODE            NOMINATED NODE    READINESS GATES
pod/aceest-app-primary-759745bdf7-4mfrc    1/1      Running   0            102s   10.244.0.96   minikube        <none>             <none>
pod/aceest-app-primary-759745bdf7-c96hl    1/1      Running   0            103s   10.244.0.94   minikube        <none>             <none>
pod/aceest-app-shadow-689697c744-zb6fz     1/1      Running   0            103s   10.244.0.95   minikube        <none>             <none>

NAME                                TYPE    CLUSTER-IP    EXTERNAL-IP    PORT(S)    AGE    SELECTOR
service/aceest-app-primary           ClusterIP  10.111.223.206 <none>         80/TCP     103s   app=aceest-app-shadow,track=primary
service/aceest-app-shadow            ClusterIP  10.98.110.250  <none>         80/TCP     103s   app=aceest-app-shadow,track=shadow

NAME                                CLASS    HOSTS                                ADDRESS    PORTS    AGE
ingress.networking.k8s.io/aceest-app-shadow  nginx    aceest-shadow.local                192.168.49.2  80      103s
Name:                                aceest-app-shadow
Labels:                               <none>
Namespace:                           aceest
Address:                              192.168.49.2
Ingress Class:                       nginx
Default backend: <default>
Rules:
Host:                                ----
----                                Path    Backends
aceest-shadow.local                  /       aceest-app-primary:80 (10.244.0.94:5000,10.244.0.96:5000)
Annotations:
  nginx.ingress.kubernetes.io/mirror-target: http://aceest-app-shadow.aceest.svc.cluster.local
  nginx.ingress.kubernetes.io/rewrite-target: /
Events:
Type    Reason    Age    From    Message
----    -
Normal  Sync      93s   (x2 over 103s)   nginx-ingress-controller    Scheduled for sync

```

```

pratheushakk@Pratheushas-MacBook-Pro aceest-app % kubectl port-forward -n ingress-nginx svc/ingress-nginx-controller 8080:80

```

```

Forwarding from 127.0.0.1:8080 -> 80
Forwarding from [::1]:8080 -> 80

```

```

pratheushakk@Pratheushas-MacBook-Pro aceest-app % ./scripts/deploy_strategy.sh ab-testing aceest aceest-app:v4.0.1 aceest-app:v4.0.0

```

```

namespace/aceest unchanged
configmap/aceest-config unchanged
service/aceest-app-a created
service/aceest-app-b created
deployment.apps/aceest-app-a created
deployment.apps/aceest-app-b created
ingress.networking.k8s.io/aceest-app-primary created
ingress.networking.k8s.io/aceest-app-experiment created
deployment.apps/aceest-app-a image updated
deployment.apps/aceest-app-b image updated
Waiting for deployment "aceest-app-a" rollout to finish: 1 out of 2 new replicas have been updated...
Waiting for deployment "aceest-app-a" rollout to finish: 1 out of 2 new replicas have been updated...
Waiting for deployment "aceest-app-a" rollout to finish: 1 out of 2 new replicas have been updated...
Waiting for deployment "aceest-app-a" rollout to finish: 1 out of 2 new replicas have been updated...
Waiting for deployment "aceest-app-a" rollout to finish: 1 old replicas are pending termination...
Waiting for deployment "aceest-app-a" rollout to finish: 1 old replicas are pending termination...
Waiting for deployment "aceest-app-a" rollout to finish: 1 old replicas are pending termination...
deployment "aceest-app-a" successfully rolled out
deployment "aceest-app-b" successfully rolled out
Send header 'X-Experiment: B' to route a client into variant B.
NAME                                READY    STATUS    RESTARTS    AGE
pod/aceest-app-a-68d8bc6cc5-65wg4    0/1      Terminating   0            2s
pod/aceest-app-a-68d8bc6cc5-djmuj    0/1      Terminating   0            2s
pod/aceest-app-a-7db78f6dcd-7nkyd    1/1      Running         0            2s
pod/aceest-app-a-7db78f6dcd-s5zvd    1/1      Running         0            1s
pod/aceest-app-b-656559c4dd-6jmmr    0/1      Terminating   0            2s
pod/aceest-app-b-656559c4dd-rs6zb    0/1      Terminating   0            2s
pod/aceest-app-b-6c7665c658-2zj5g    1/1      Running         0            1s
pod/aceest-app-b-6c7665c658-q75sc    1/1      Running         0            2s
pod/aceest-app-primary-759745bdf7-4mfrc    1/1      Running         0            4m45s
pod/aceest-app-primary-759745bdf7-c96hl    1/1      Running         0            4m46s
pod/aceest-app-shadow-689697c744-zb6fz    1/1      Running         0            4m46s

NAME                                TYPE    CLUSTER-IP    EXTERNAL-IP    PORT(S)    AGE
service/aceest-app-a                 ClusterIP  10.101.57.108  <none>         80/TCP     2s
service/aceest-app-b                 ClusterIP  10.106.12.209  <none>         80/TCP     2s
service/aceest-app-primary           ClusterIP  10.111.223.206 <none>         80/TCP     4m46s
service/aceest-app-shadow            ClusterIP  10.98.110.250  <none>         80/TCP     4m46s

NAME                                CLASS    HOSTS                                ADDRESS    PORTS    AGE
ingress.networking.k8s.io/aceest-app-experiment  nginx    aceest-ab.local                80      2s
ingress.networking.k8s.io/aceest-app-primary     nginx    aceest-ab.local                80      2s
ingress.networking.k8s.io/aceest-app-shadow      nginx    aceest-shadow.local            192.168.49.2  80      4m46s

```

```
pratheushakk@Pratheushas-MacBook-Pro aceest-app % curl -H "Host: aceest-ab.local" http://127.0.0.1:8080/
curl -H "Host: aceest-ab.local" -H "X-Experiment: B" http://127.0.0.1:8080/
```

```
{
  "message": "ACEest Fitness & Gym API is running"
}
{
  "message": "ACEest Fitness & Gym API is running"
}
```

```
pratheushakk@Pratheushas-MacBook-Pro aceest-app % kubectl rollout history deployment/aceest-app-primary -n aceest
kubectl rollout history deployment/aceest-app-shadow -n aceest
```

```
deployment.apps/aceest-app-primary
REVISION  CHANGE-CAUSE
1          <none>
2          Primary shadow environment deployed with stable image aceest-app:v4.0.0

deployment.apps/aceest-app-shadow
REVISION  CHANGE-CAUSE
1          <none>
2          Shadow environment deployed with candidate image aceest-app:v4.0.1
```

7.5 A/B TESTING

```
pratheushakk@Pratheushas-MacBook-Pro aceest-app % kubectl delete ns aceest --ignore-not-found=true
kubectl wait --for=delete ns/aceest --timeout=120s || true
./scripts/deploy_strategy.sh ab-testing aceest aceest-app:v4.0.1 aceest-app:v4.0.0
```

```
namespace "aceest" deleted
namespace/aceest created
configmap/aceest-config created
service/aceest-app-a created
service/aceest-app-b created
deployment.apps/aceest-app-a created
deployment.apps/aceest-app-b created
ingress.networking.k8s.io/aceest-app-primary created
ingress.networking.k8s.io/aceest-app-experiment created
deployment.apps/aceest-app-a image updated
deployment.apps/aceest-app-b image updated
Waiting for deployment "aceest-app-a" rollout to finish: 1 out of 2 new replicas have been updated...
Waiting for deployment "aceest-app-a" rollout to finish: 1 out of 2 new replicas have been updated...
Waiting for deployment "aceest-app-a" rollout to finish: 1 out of 2 new replicas have been updated...
Waiting for deployment "aceest-app-a" rollout to finish: 1 out of 2 new replicas have been updated...
Waiting for deployment "aceest-app-a" rollout to finish: 1 out of 2 new replicas have been updated...
Waiting for deployment "aceest-app-a" rollout to finish: 1 old replicas are pending termination...
Waiting for deployment "aceest-app-a" rollout to finish: 1 old replicas are pending termination...
Waiting for deployment "aceest-app-a" rollout to finish: 1 old replicas are pending termination...
Waiting for deployment "aceest-app-a" rollout to finish: 1 old replicas are pending termination...
deployment "aceest-app-a" successfully rolled out
deployment "aceest-app-b" successfully rolled out
Send header 'X-Experiment: B' to route a client into variant B.
```

NAME	READY	STATUS	RESTARTS	AGE
pod/aceest-app-a-68d8bc6cc5-k8z97	0/1	Terminating	0	4s
pod/aceest-app-a-7db78f6dcd-4g87p	1/1	Running	0	2s
pod/aceest-app-a-7db78f6dcd-zkfr9	1/1	Running	0	4s
pod/aceest-app-b-6c7665c658-gjt27	1/1	Running	0	4s
pod/aceest-app-b-6c7665c658-jm4t9	1/1	Running	0	2s

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)	AGE
service/aceest-app-a	ClusterIP	10.107.177.206	<none>	80/TCP	5s
service/aceest-app-b	ClusterIP	10.106.241.197	<none>	80/TCP	5s

NAME	CLASS	HOSTS	ADDRESS	PORTS	AGE
ingress.networking.k8s.io/aceest-app-experiment	nginx	aceest-ab.local		80	4s
ingress.networking.k8s.io/aceest-app-primary	nginx	aceest-ab.local		80	4s

```
pratheushakk@Pratheushas-MacBook-Pro aceest-app % kubectl annotate deployment/aceest-app-a -n aceest \
  kubernetes.io/change-cause="Variant A deployed with stable image aceest-app:v4.0.0" --overwrite
```

```
kubectl annotate deployment/aceest-app-b -n aceest \
  kubernetes.io/change-cause="Variant B deployed with candidate image aceest-app:v4.0.1" --overwrite
```

```
deployment.apps/aceest-app-a annotated
deployment.apps/aceest-app-b annotated
```

```

pratheushakk@Pratheushas-MacBook-Pro aceest-app % kubectl set image deployment/aceest-app-a aceest-app=aceest-app:v4.0.0 -n aceest
kubectl set image deployment/aceest-app-b aceest-app=aceest-app:v4.0.1 -n aceest
kubectl rollout status deployment/aceest-app-a -n aceest
kubectl rollout status deployment/aceest-app-b -n aceest
kubectl get deploy,pods,svc,ingress -n aceest -o wide
kubectl describe ingress aceest-app-experiment -n aceest

deployment "aceest-app-a" successfully rolled out
deployment "aceest-app-b" successfully rolled out
NAME                READY    UP-TO-DATE    AVAILABLE    AGE    CONTAINERS    IMAGES                SELECTOR
deployment.apps/aceest-app-a    2/2      2              2            22s    aceest-app    aceest-app:v4.0.0    app=aceest-app-ab,strategy=ab-testing,variant=a
deployment.apps/aceest-app-b    2/2      2              2            22s    aceest-app    aceest-app:v4.0.1    app=aceest-app-ab,strategy=ab-testing,variant=b

NAME                READY    STATUS    RESTARTS    AGE    IP            NODE            NOMINATED NODE    READINESS GATES
pod/aceest-app-a-7db78f6dcd-4g87p    1/1      Running    0           20s    10.244.0.111   minikube        <none>             <none>
pod/aceest-app-a-7db78f6dcd-zkfr9    1/1      Running    0           22s    10.244.0.109   minikube        <none>             <none>
pod/aceest-app-b-6c7665c658-gjt27    1/1      Running    0           22s    10.244.0.110   minikube        <none>             <none>
pod/aceest-app-b-6c7665c658-jm4t9    1/1      Running    0           20s    10.244.0.112   minikube        <none>             <none>

NAME                TYPE        CLUSTER-IP    EXTERNAL-IP    PORT(S)    AGE    SELECTOR
service/aceest-app-a    ClusterIP    10.107.177.206    <none>         80/TCP     23s    app=aceest-app-ab,variant=a
service/aceest-app-b    ClusterIP    10.106.241.197    <none>         80/TCP     23s    app=aceest-app-ab,variant=b

NAME                CLASS    HOSTS                ADDRESS        PORTS    AGE
ingress.networking.k8s.io/aceest-app-experiment    nginx    aceest-ab.local      192.168.49.2   80       22s
ingress.networking.k8s.io/aceest-app-primary        nginx    aceest-ab.local      192.168.49.2   80       22s
Name:                aceest-app-experiment
Labels:               <none>
Namespace:            aceest
Address:              192.168.49.2
Ingress Class:        nginx
Default backend:      <default>
Rules:
  Host                Path    Backends
  ----                -
  aceest-ab.local      /       aceest-app-b:80 (10.244.0.110:5000,10.244.0.112:5000)
Annotations:
  nginx.ingress.kubernetes.io/canary: true
  nginx.ingress.kubernetes.io/canary-by-header: X-Experiment
  nginx.ingress.kubernetes.io/canary-by-header-value: B
Events:
  Type    Reason    Age    From    Message
  ----    -
  Normal  Sync      9s    (x2 over 22s)    nginx-ingress-controller    Scheduled for sync

```

```

pratheushakk@Pratheushas-MacBook-Pro aceest-app % curl -H "Host: aceest-ab.local" http://127.0.0.1:8080/
curl -H "Host: aceest-ab.local" -H "X-Experiment: B" http://127.0.0.1:8080/

{
  "message": "ACEest Fitness & Gym API is running"
}
{
  "message": "ACEest Fitness & Gym API is running"
}

```

```

pratheushakk@Pratheushas-MacBook-Pro aceest-app % kubectl rollout history deployment/aceest-app-a -n aceest
kubectl rollout history deployment/aceest-app-b -n aceest

deployment.apps/aceest-app-a
REVISION  CHANGE-CAUSE
1         <none>
2         Variant A deployed with stable image aceest-app:v4.0.0

deployment.apps/aceest-app-b
REVISION  CHANGE-CAUSE
1         <none>
2         Variant B deployed with candidate image aceest-app:v4.0.1

```

8. KEY AUTOMATION OUTCOMES

- End-to-end CI/CD workflow implemented successfully across GitHub Actions, Jenkins, Docker, Docker Hub container Registry, and Kubernetes/Minikube.
- Automated Pytest/Selenium validation and Sonar CLI static analysis improved delivery confidence before image promotion.
- The secure multi-stage Docker image improved consistency across local verification, Jenkins, Docker Hub, and Kubernetes/Minikube execution..

- Rolling update, blue-green, canary, shadow, and A/B testing evidence improved release safety and rollback readiness.

9. CHALLENGES AND MITIGATIONS

9.1 ASSIGNMENT 1

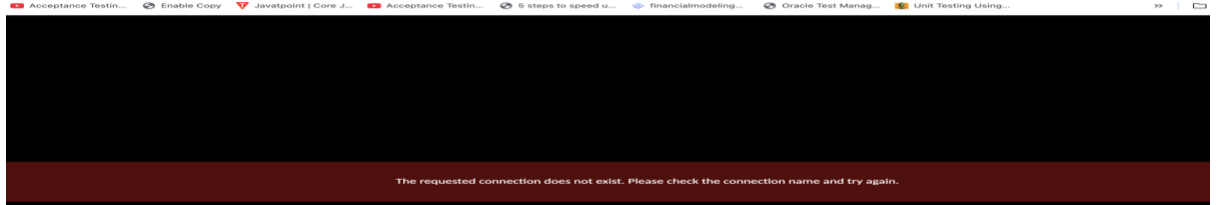
- **Challenge 1:** Jenkins and application endpoints could not be exposed reliably outside the Azure virtual lab network.
- **Mitigation:** Jenkins was kept on localhost and configured with Poll SCM so it could monitor GitHub without a public webhook endpoint.
- **Challenge 2:** A full SonarQube server UI was not practical because the lab environment restricted access to the required external port.
- **Mitigation:** Sonar CLI was used directly in the Jenkins pipeline with sonar-project.properties, and the analysis output was archived as evidence.
- **Challenge 3:** Private container deployment required Azure Web App to pull images from JFrog Artifactory securely.
- **Mitigation:** Jenkins stores the Azure Service Principal JSON and JFrog registry credentials securely, then configures Azure Web App for Containers to pull the private image.

9.2 ASSIGNMENT 2

- **Challenge 1:** The DevOps lab facility provided for the assignment was available only for 24 hours. After the lab access expired, the original hosted lab environment could no longer be used, so the CI/CD setup had to be recreated on the local machine.

A fresh AWS instance was launched for you. Please wait for a few minutes before connecting. ✕

Course (Virtual Lab)	Machine	Time Used	Time Remaining	State	Control
25S2NSP3-Intro to DevOps(Virtual Lab)	i-090907bb765d82fef	22h 11m		saved	Start



Mitigation: Jenkins, SonarQube, Docker, and Minikube were configured on the local machine to continue the assignment work after the DevOps lab facility expired. This allowed the pipeline stages and deployment flow to be demonstrated without depending on the temporary lab environment.

- **Challenge 2:** The original flow in Assignment1 expected an artifact/image registry such as JFrog, but the free option was no longer available.

Mitigation: Docker Hub was used as the container image registry because it provides a suitable free option for this assignment. Jenkins credentials were configured securely using dockerhub-credentials.

- **Challenge 3:** Upgrading the Azure account could introduce paid resources. To avoid unexpected billing, Azure deployment was removed from the local Jenkins pipeline and kept only as an optional future remote deployment activity.

Mitigation: The local Jenkins pipeline was updated to focus on local CI/CD validation only: build, test, Docker image creation, Docker Hub push, Minikube deployment, and rollback.

- **Challenge 4:** The Azure free tier setup used in Assignment 1 has expired. Because of this, the application is no longer available at the Azure endpoint submitted for Assignment 1.

Mitigation: Since the Azure free tier endpoint is no longer active, Minikube was used as the Kubernetes deployment target. This allowed the complete deployment flow to be demonstrated locally without cloud cost.

10. CONCLUSION

This assignment demonstrates how the ACEest Fitness & Gym application can move from code change to tested artifact, registry publication, Minikube/Kubernetes deployment, Kubernetes rollout validation, and rollback using controlled DevOps automation.